



**SYDNEY BOYS HIGH SCHOOL**  
MOORE PARK, SURRY HILLS

**2000**

**TRIAL HIGHER SCHOOL  
CERTIFICATE EXAMINATION**

# Mathematics      Extension 1

*Examiner: B. Dowdell*

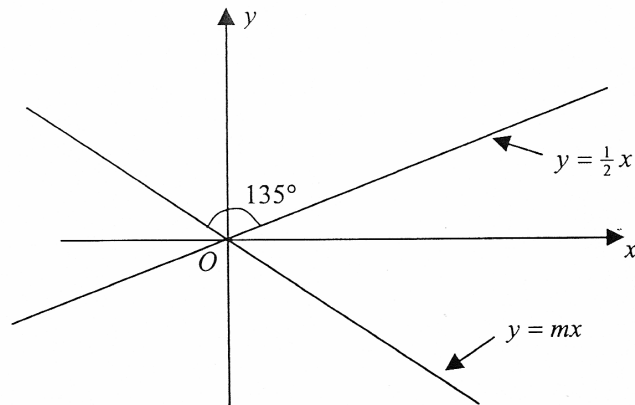
- (a) State the domain and range of  $4\sin^{-1} 3x$  2
- (b) Solve for  $x$ :  $(x - 2)^2 \leq 4$  2
- (c) Differentiate: 4
- (i)  $x \cos^{-1} 2x$
- (ii)  $\frac{1}{4 + x^2}$
- (d) Find  $x$  correct to 3 decimal places if  $x^{\frac{3}{4}} = 10$  2
- (e) The point  $P(11, 7)$  divides  $AB$  externally in the ratio 3:1. If  $B$  is  $(6, 5)$ , find the coordinates of  $A$ . 2

Question 2: START A NEW BOOKLET

Marks

(a)

2



The angle between the lines  $y = mx$  and  $y = \frac{1}{2}x$  is  $135^\circ$ . Find the exact value of  $m$ .

(b) Using  $u = \sqrt{x}$  evaluate  $\int_1^4 \frac{dx}{x + \sqrt{x}}$

2

(c) Write down the exact value of  $\cos^{-1}(\cos \frac{4\pi}{3})$

2

(d) Find a primitive of

4

(i)  $\frac{2}{\sqrt{1-4x^2}}$

(ii)  $\frac{x}{4+x^2}$

(e) Find the values of  $a$  for which  $f(x) = e^{-ax}(x-a)$  is stationary at  $x = \frac{5}{2}$ .

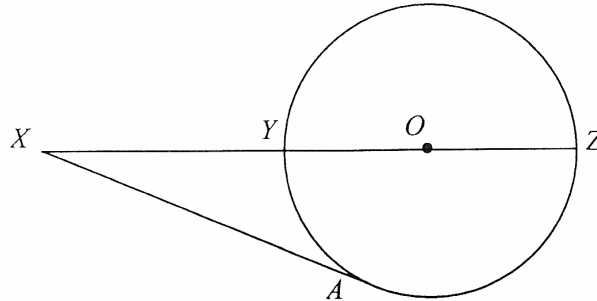
2

Question 3: START A NEW BOOKLET

Marks

(a)

3



$O$  is the centre of the circle,  $XA$  is a tangent.

$$XY = 3 \text{ and } XA = 5$$

Calculate the size of  $\angle AXY$  correct to the nearest minute.

- (b) (i) Sketch the graphs of  $y = e^x$  and  $y = \cos x$  on the same diagram for  $0 \leq x \leq \frac{\pi}{2}$ , clearly showing any points of intersection.

4

Shade the area enclosed by the two curves and the line  $x = \frac{\pi}{2}$ .

- (ii) Calculate the volume of the solid formed when this area is rotated about the  $x$  axis.

- (c) (i) Prove that  $\frac{d^2x}{dt^2} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$ .

5

- (ii) A particle moves in a straight line with velocity given by  $v^2 = 36 - 4x^2$  where  $x$  is measured in metres and is the displacement from a fixed point  $O$  and  $t$  is the time measured in seconds.

( $\alpha$ ) Show that the motion is simple harmonic

( $\beta$ ) Find the period and amplitude of the motion.

**Question 4: START A NEW BOOKLET**

- (a) When  $P(x) = ax^3 + bx + c$  is divided by  $x - 1$  the remainder is  $-4$ . 3  
When  $P(x)$  is divided by  $x^2 - 4$ , the remainder is  $-4x + 3$ .  
Find  $a, b$  and  $c$ .
- (b) Prove by induction that 4  
$$1 + (1 + 2) + (1 + 2 + 3) + \dots + (1 + 2 + 3 + \dots + n) = \frac{n}{6}(n + 1)(n + 2)$$
for all positive integers  $n$ .
- (c) (i) Show that the point  $A(6p, 3p^2)$  lies on the parabola  $x^2 = 12y$ . 5  
(ii) The chord joining  $A(6p, 3p^2)$  and  $B(6q, 3q^2)$ , when produced, passes through  $C(8, 0)$ . Show that  $4(p + q) = 3pq$  and hence find the locus of  $M$ , the midpoint of  $AB$ .

**Question 5: START A NEW BOOKLET**

Marks

(a) Prove that  $2 \tan^{-1} \theta = \tan^{-1} \left( \frac{2\theta}{1-\theta^2} \right)$  provided that  $|\theta| < 1$ .

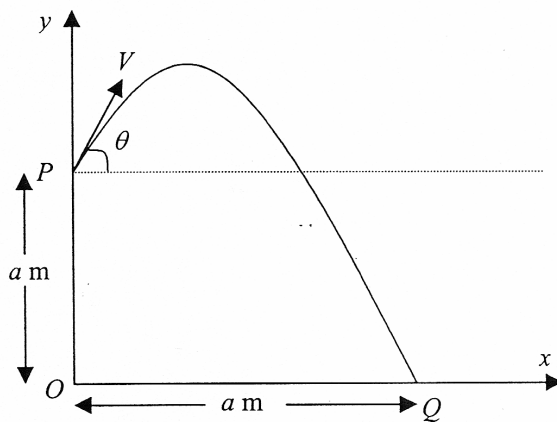
2

(b) A balloon is being filled with helium at a constant rate of  $30 \text{ cm}^3/\text{s}$ . Find the rate at which the surface area is increasing when its diameter is 40 cm.

4

(c)

6



A projectile is fired from a point  $P$ ,  $a$  metres above  $O$  with an initial velocity  $V \text{ ms}^{-1}$  at an angle of elevation of  $\theta$ . It is subject to a constant downward acceleration of  $g \text{ ms}^{-2}$ .

- (i) Find expressions for the horizontal ( $x$ ) and vertical ( $y$ ) displacements from  $P$  after  $t$  seconds.
- (ii) Show that the time taken to reach  $Q$ ,  $a$  metres from  $O$  in a horizontal direction is given by  $\frac{2V(\sin \theta + \cos \theta)}{g}$  seconds.
- (iii) Show that  $a = \frac{V^2(\sin 2\theta + \cos 2\theta + 1)}{g}$  metres.

**Question 6: START A NEW BOOKLET**

- ) Eight people attend a meeting. They are provided with two circular tables, one seating 3 people, the other 5 people. 4
- (i) How many seating arrangements are possible?
- (ii) If the seating is done randomly, what is the probability that a particular couple are on different tables?

- ) If  $f(x) = u(x) - \ln(u(x) + 1)$  4
- (i) Show that  $f'(x) = \frac{u(x).u'(x)}{1 + u(x)}$ .
- (ii) Hence or otherwise evaluate

$$\int_0^{\frac{\pi}{2}} \frac{\cos x \cdot \sin x}{1 + \sin x} dx$$

- ) A function  $L(x)$  is defined by 4

$$L(x) = Pe^{\frac{x}{3}} + Qe^{-\frac{2x}{3}} \text{ where } P \text{ and } Q \text{ are constants.}$$

It is given that  $L(0) = 30$  and  $L'(0) = -14$ .

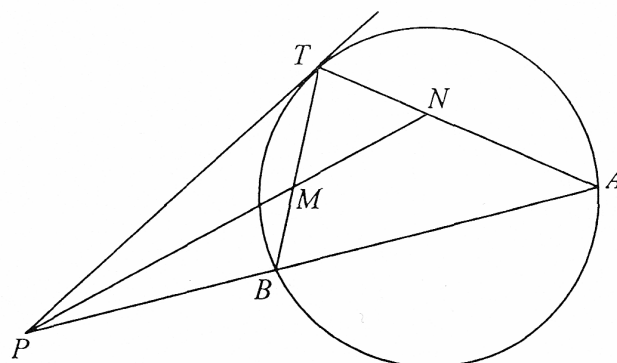
- (i) Find the values of  $P$  and  $Q$ .
- (ii) Find  $L'(3)$  and explain why  $L(x)$  must have a minimum for some value of  $x$  between 0 and 3.

Question 7: START A NEW BOOKLET

Mark

(a)

3



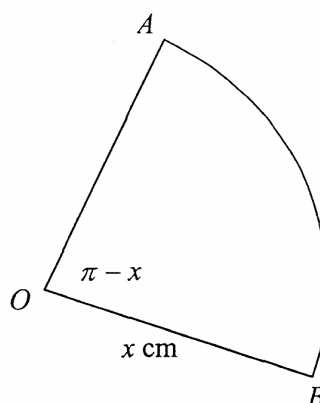
$AB$  is any chord of a circle.  $AB$  is produced to  $P$ , and  $PT$  is a tangent. The bisector of  $\angle APT$  meets  $TB$  at  $M$  and  $TA$  at  $N$ .

- (i) Copy the diagram into your answer booklet.
- (ii) Prove that  $\triangle TMN$  is isosceles.

(b)

9

$AOB$  is a sector of a circle, such that, when the radius is  $x$  cm,  $\angle AOB = (\pi - x)$  radians and  $x$  varies from 0 to  $\pi$ .



- (i) Find the maximum value of the perimeter of sector  $AOB$ . Comment on the minimum value of the perimeter of the sector.
- (ii) If the area of **triangle**  $AOB$  is given by  $t(x)$ 
  - ( $\alpha$ ) Show that  $t(x) = \frac{x^2 \sin x}{2}$ .
  - ( $\beta$ ) Show that when  $t(x)$  is a maximum,  $2 \tan x = -x$ .
  - ( $\gamma$ ) By sketching  $y = \tan x$  and a suitable line, show that a solution to the equation in ( $\beta$ ) is close to  $x = \frac{3\pi}{4}$ .
  - ( $\delta$ ) Taking  $\frac{3\pi}{4}$  as a first approximation, use Newton's method once to obtain a better approximation (leave your answer in terms of  $\pi$ ).

END OF PAPER





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**Mathematics    Extension 1**

**Sample Solutions**

(Q1) (a)  $y = 4 \sin^{-1} 3x$

$$D: -1 \leq 3x \leq 1 \quad R: -\frac{\pi}{2} \leq \frac{y}{4} \leq \pi/2$$

$$-\frac{1}{3} \leq x \leq \frac{1}{3} \quad -2\pi \leq y \leq 2\pi$$

(b)  $(x-2)^2 \leq 4$

$$\circ \quad -2 \leq x-2 \leq 2$$

$$0 \leq x \leq 4$$

$$\begin{aligned} \text{(c)} \quad \text{ii)} \quad \frac{d(\cos^{-1} 2x)}{dx} &= \cos^{-1} 2x - x \times \frac{2}{\sqrt{1-4x^2}} \\ &= \cos^{-1} x - \frac{2x}{\sqrt{1-4x^2}} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad d\left(\frac{1}{4+x^2}\right) &= \frac{d(4+x^2)^{-1}}{dx} \\ &= -(4+x^2)^{-2} \times 2x \\ &= \frac{-2x}{(4+x^2)^2} \end{aligned}$$

$$\begin{aligned} \text{(d)} \quad x^{3/4} &= 10 \\ \therefore x &= 10^{4/3} \\ &\approx 21.544 \end{aligned}$$

$$e) \quad P\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$

$$A(x_1, y_1) \quad B(6, 5) \quad P(11, 7) \quad \begin{matrix} m & n \\ 3 & -1 \end{matrix}$$

$$\therefore 11 = \frac{3 \times 6 - x_1}{2}, \quad 7 = \frac{3 \times 5 - y_1}{2}$$

$$18 - x_1 = 22, \quad 15 - y_1 = 14$$

$$x_1 = -4, \quad y_1 = 1$$

$$A(-4, 1)$$

(2) (a) the acute angle is  $45^\circ$

$$\tan 45 = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \quad m_2 = \frac{1}{2}$$

$$\therefore \tan 45^\circ = \left| \frac{m_1 - \frac{1}{2}}{1 + m_1 \times \frac{1}{2}} \right|$$

$$\therefore \left| \frac{m_1 - \frac{1}{2}}{1 + \frac{m_1}{2}} \right| = 1 \Rightarrow \left| \frac{m_1 - \frac{1}{2}}{2 + m_1} \right| = 2$$

$$\therefore \frac{m - \frac{1}{2}}{2 + m} = 2, \quad \frac{m - \frac{1}{2}}{2 + m} = -2$$

$$m - \frac{1}{2} = 4 + 2m \quad m - \frac{1}{2} = -4 + 2m$$

$$m = 4\frac{1}{2}, \quad m = -3\frac{1}{2}$$

$$\therefore \boxed{m = -3\frac{1}{2}, 4\frac{1}{2}}$$

(b)  $u = \sqrt{x} \Rightarrow x = u^2$

$$\therefore dx = 2u du$$

$$\int_1^4 \frac{dx}{x + \sqrt{x}}$$

$$x=1 \Rightarrow u=1$$

$$x=4 \Rightarrow u=2$$

$$= \int_1^2 \frac{2u du}{u^2 + u}$$

$$= \int_1^2 \frac{2}{u+2}$$

$$= \ln|u+2| \Big|_1^2$$

$$= \ln 4 - \ln 3$$

$$= \ln\left(\frac{4}{3}\right)$$

(c)  $\cos^{-1}(\cos \frac{4\pi}{3})$

$$= \cos^{-1}\left(-\frac{1}{2}\right)$$

$$= \frac{2\pi}{3}$$

(d) (i)  $\int \frac{2 dx}{\sqrt{1-4x^2}} = \sin^{-1}(2x) + C$

(ii)  $\int \frac{x}{4+x^2} dx$

$$= \frac{1}{2} \int \frac{2x}{4+x^2} dx$$

$$= \frac{1}{2} \ln(x^2+4) + C$$

(2) (e)

$$f(x) = e^{-ax}(x-a)$$

$$f'(x) = e^{-ax} + (x-a) \times -ae^{-ax}$$

$$= e^{-ax}(1 - a(x-a))$$

$$f'(\frac{5}{2}) = 0$$

$$e^{-ax} \neq 0 \quad \therefore 1 - a(\frac{5}{2} - a) = 0$$

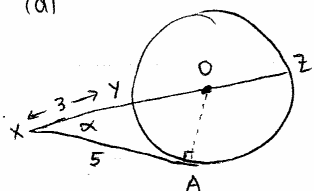
$$\therefore 2 - 5a + 2a^2 = 0$$

$$\therefore 2a^2 - 5a + 2 = 0$$

$$(2a-1)(a-2) = 0$$

$$a = \frac{1}{2}, 2$$

(3) (a)



$$XZ \cdot XY = XA^2$$

$$\therefore 25 = 3 \times XZ$$

$$XZ = \frac{25}{3} = 8\frac{1}{3}$$

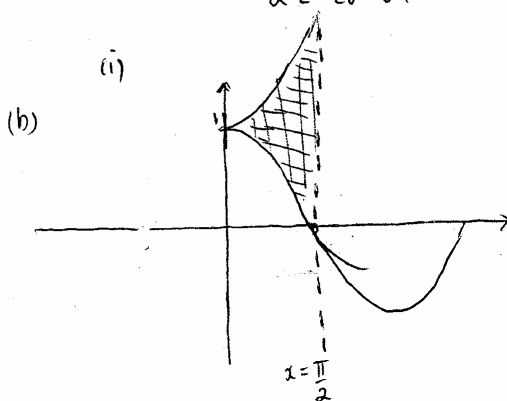
$$\therefore YZ = 5\frac{1}{3}$$

$$\therefore OA = 8/3$$

$$\text{Let } \alpha = \angle AXY$$

$$\tan \alpha = \frac{8/3}{5} = 8/15$$

$$\alpha = 28^\circ 04'$$



$$\text{Area} = \int_0^{\pi/2} (e^x - \cos x) dx$$

$$= [e^x - \sin x]_0^{\pi/2}$$

$$= (e^{\pi/2} - \sin \frac{\pi}{2}) - (e^0 - \sin 0)$$

$$= e^{\pi/2} - 1 - 1$$

$$= e^{\pi/2} - 2$$

3 (b) (ii)

$$V = \pi \int_0^{\pi/2} (e^{2x} - \cos^2 x) dx$$

$$\left[ \cos^2 x = \frac{1}{2} (1 + \cos 2x) \right]$$

$$= \pi \int_0^{\pi/2} \left( e^{2x} - \frac{1}{2} - \frac{1}{2} \cos 2x \right) dx$$

$$= \pi \left[ \frac{1}{2} e^{2x} - \frac{1}{2} x - \frac{1}{4} \sin 2x \right]_0^{\pi/2}$$

$$= \pi \left[ \left( \frac{1}{2} e^{\pi} - \frac{\pi}{4} \right) - \left( \frac{1}{2} \right) \right]$$

$$= \frac{\pi}{2} \left( e^{\pi} - \frac{\pi}{2} - 1 \right)$$

(c) (i)  $RHS = d\left(\frac{1}{2}v^2\right)$

$$= d\left(\frac{1}{2}v^2\right) \times \frac{dv}{dx}$$

$$= v \frac{dv}{dx} = \frac{dx}{dt} \cdot \frac{dv}{dx}$$

$$= \frac{dv}{dt}$$

$$= a$$

$$= \ddot{x}$$

$$= LHS$$

(ii)  $v^2 = 36 - 4x^2$

$$\frac{1}{2}v^2 = 18 - x^2 \Rightarrow a = \frac{d\left(\frac{1}{2}v^2\right)}{dx}$$

(x)  $a = -2x$

This is one of the defining equations for SHM, centred at  $x=0$

(p)  $n^2 = 2 \Rightarrow n = \sqrt{2}$

$$T = \frac{2\pi}{n} = \frac{2\pi}{\sqrt{2}} = \sqrt{2}\pi$$

$$v=0 \Rightarrow x^2 = 18$$

$$x = \pm 3\sqrt{2}$$

$$\therefore \text{Amplitude} = 3\sqrt{2} \text{ m}$$

$$(4) (a) \quad P(x) = ax^3 + bx^2 + c$$

$$P(1) = -4$$

$$\Rightarrow a + b + c = -4 \quad - (1)$$

$$P(x) = (x^2 - 4)Q(x) + (-4x + 3)$$

$$\therefore P(2) = -5$$

$$\Rightarrow 8a + 2b + c = -5 \quad - (2)$$

$$P(-2) = 11$$

$$\Rightarrow -8a - 2b + c = 11 \quad - (3)$$

$$(2) + (3): \quad 2c = 6$$

$$c = 3$$

$$\begin{array}{l} (1) \Rightarrow a + b = -7 \quad - (4) \\ (2) \Rightarrow 8a + 2b = -8 \quad - (5) \\ \quad \quad 4a + b = -4 \quad - (6) \end{array} \quad \left. \begin{array}{l} - \\ - \end{array} \right\}$$

$$(6) - (4) \quad 3a = 3$$

$$a = 1$$

$$\text{sub into } (4) \quad b = -8$$

$$\therefore a = 1, b = -8, c = 3$$

$$(b) \quad 1 + (1+2) + (1+2+3) + \dots + (1+2+3+\dots+n) = \frac{n}{6}(n+1)(n+2), \quad n > 0$$

Using the sum of an arithmetic series

$$\therefore 1 + (1+2) + \dots + \frac{n(n+1)}{2} = \frac{n}{6}(n+1)(n+2), \quad n > 0 \quad - (*)$$

$$\text{Test } n=1: \quad \text{LHS} = 1$$

$$\text{RHS} = \frac{1}{6}(2)(3) = 1$$

$$\therefore \text{true for } n=1$$

Assume (\*) is true for some integer  $n=k$ .

$$\therefore 1 + (1+2) + \dots + \frac{k(k+1)}{2} = \frac{k}{6}(k+1)(k+2)$$

We need to prove (\*) is true for the integer  $n=k+1$

$$\text{i.e. } 1 + (1+2) + \dots + \frac{(k+1)(k+2)}{2} = \frac{(k+1)(k+2)(k+3)}{6}$$

4(b)

$$LHS = 1 + (1+2) + \dots + \frac{k(k+1)}{2} + \frac{(k+1)(k+2)}{2}$$

$$= \frac{k(k+1)(k+2)}{6} + \frac{(k+1)(k+2)}{2}$$

$$= (k+1)(k+2) \left[ \frac{k}{6} + \frac{1}{2} \right]$$

$$= (k+1)(k+2) \frac{(k+3)}{6}$$

$$= \frac{1}{6} (k+1)(k+2)(k+3)$$

$$= RHS$$

$\therefore$  Since the statement is true for  $n=k+1$  when the statement is true for  $n=k$ . By the principle of mathematical induction

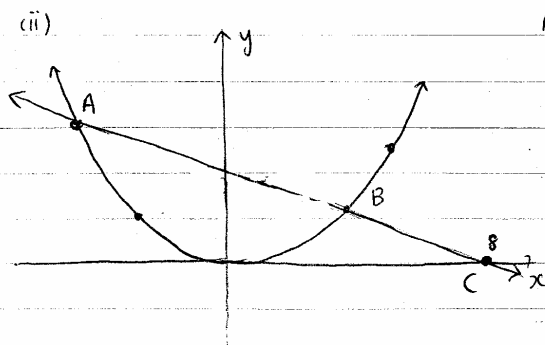
$$1 + (1+2) + \dots + (1+2+\dots+n) = \frac{n(n+1)(n+2)}{6}, \quad n > 0$$

(c) (i)  $A(6p, 3p^2)$

$$LHS = x^2 = 36p^2$$

$$RHS = 12y = 12(3p^2) = 36p^2$$

$\therefore A$  lies on  $x^2 = 12y$



$$\begin{matrix} x_1 & y_1 & x_2 & y_2 \\ A(6p, 3p^2) & B(6q, 3q^2) \end{matrix}$$

$$m_{AB} = \frac{3q^2 - 3p^2}{6q - 6p} = \frac{3(q-p)(q+p)}{6(q-p)} = \frac{q+p}{2}$$

$$\therefore y - 3q^2 = \frac{q+p}{2}(x - 6q)$$

$$2y - 6q^2 = (q+p)x - 6q(q+p)$$

$$2y = (q+p)x - 6qp \quad \text{--- (1)}$$

4 (c) (ii)  $C(8,0)$  lies on (1)

$$\text{i.e. } 0 = (q+p)8 - 6qp$$

$$\therefore 6qp = 8(q+p) \Rightarrow 3pq = 4(p+q) \quad - (*)$$

$$\text{Midpoint AB } \left( \frac{6p+6q}{2}, \frac{3p^2+3q^2}{2} \right)$$

$$x = 3(p+q) \quad y = \frac{3}{2}(p^2+q^2)$$

$$= \frac{3}{2}[(p+q)^2 - 2pq]$$

$$= \frac{3}{2}(p+q)^2 - 3pq$$

$$= \frac{3}{2}(p+q)^2 - 4(p+q) \quad \text{from } (*)$$

$$= \frac{3}{2}\left[\frac{x}{3}\right]^2 - 4\left[\frac{x}{3}\right]$$

$$= \frac{x^2}{6} - \frac{4x}{3}$$

$$\therefore \text{Locus of M is } y = \frac{x^2}{6} - \frac{4x}{3}$$



Question 5(a)

$$2 \tan^{-1} \theta = \tan^{-1} \left( \frac{2\theta}{1-\theta^2} \right)$$

$|\theta| < 1$

$$\begin{aligned} \tan(2 \tan^{-1} \theta) &= \frac{2 \tan(\tan^{-1} \theta)}{1 - \tan^2(\tan^{-1} \theta)} \\ &= \frac{2\theta}{1-\theta^2} \end{aligned}$$

$$\therefore 2 \tan^{-1} \theta = \tan^{-1} \left( \frac{2\theta}{1-\theta^2} \right)$$

Now if  $|\theta| > 1$

$$2 \tan^{-1} \theta > \frac{\pi}{2} \text{ if } \theta > 1$$

$$\text{and } 2 \tan^{-1} \theta < -\frac{\pi}{2} \text{ if } \theta < -1$$

$$\text{But } -\frac{\pi}{2} < \tan^{-1} x < \frac{\pi}{2}$$

So R.H.S. has

$$-\frac{\pi}{2} < \tan^{-1} \left( \frac{2\theta}{1-\theta^2} \right) < \frac{\pi}{2}$$

So no valid solution

$$(b) \frac{dv}{dt} = 30 \quad (v = \frac{4}{3} \pi r^3)$$

$$\frac{dv}{dt} = \frac{dv}{dr} \times \frac{dr}{dt} \quad (1)$$

$$\therefore 30 = 4\pi r^2 \times \frac{dr}{dt}$$

$$\Rightarrow \frac{dr}{dt} = \frac{15}{2\pi r^2} \quad (2)$$

$$S = 4\pi r^2$$

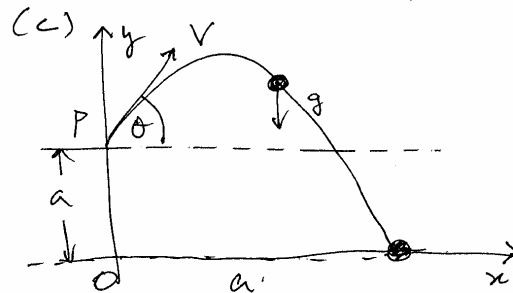
$$\frac{ds}{dt} = 8\pi r \frac{dr}{dt} \quad (3)$$

Subst (2) into (3)

$$= \cancel{8\pi r} \times \frac{15}{\cancel{2\pi r^2}}$$

$$\frac{ds}{dt} = \frac{60}{r}$$

$$\text{When } r = 20, \frac{ds}{dt} = 3$$



$$\ddot{x} = 0, \quad \dot{x} = v \cos \theta$$

$$x = (v \cos \theta) t \quad (1)$$

$$\ddot{y} = -g, \quad \dot{y} = (v \sin \theta) - gt$$

$$y = (v \sin \theta) t - \frac{gt^2}{2} + a \quad (2)$$

$$\text{When } x = a, \quad y = 0$$

$$\text{When } x = a$$

$$t = \frac{a}{v \cos \theta} \quad (3)$$

and

$$\text{When } t = \frac{a}{v \cos \theta}, \quad y = 0$$

Subst. (3) into (2)

We have

$$0 = v \sin \theta \left( \frac{a}{v \cos \theta} \right) - \frac{g}{2} \left( \frac{a}{v \cos \theta} \right)^2 + a$$

divide each term by  $a$  and rearrange.

$$0 = \tan \theta - \frac{gt}{2v \cos \theta} + 1$$

$$\frac{gt}{2v \cos \theta} = \frac{\sin \theta + \cos \theta}{\cos \theta}$$

$$\therefore t = \frac{2v(\sin \theta + \cos \theta)}{g} \quad (4)$$

$a = (v \cos \theta) t$  — (5)  
 In bst (4) into (5) we have

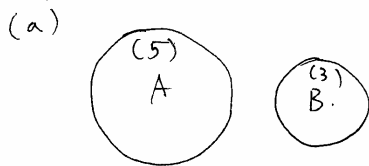
$$a = \frac{(v \cos \theta)(2v)(\sin \theta + \cos \theta)}{g}$$

$$= \frac{v^2(2\sin \theta \cos \theta + 2\cos^2 \theta)}{g}$$

$$= \frac{v^2(2\sin \theta \cos \theta + (2\cos^2 \theta - 1) + 1)}{g}$$

$$= \frac{v^2(\sin 2\theta + \cos 2\theta + 1)}{g}$$

Q nest in (6)



(a) The 1st person has 8 choices, the 2nd person has 7 choices...

$$\therefore \frac{8!}{5! \cdot 3!}$$

(ii)



$$P(E) = \frac{\frac{(2 \times 6!)}{5! \cdot 3!}}{\frac{8!}{5! \cdot 3!}}$$

$$= \frac{2}{8 \times 7} = \frac{1}{28}$$

(b) (i)  $f(x) = u(x) - \ln[u(x) + 1]$

$$f'(x) = u'(x) - \frac{u'(x)}{u(x) + 1}$$

$$= u'(x) \left[ 1 - \frac{1}{u(x) + 1} \right]$$

$$= u'(x) \left[ \frac{u(x) + 1 - 1}{u(x) + 1} \right]$$

(ii)  $\int_0^{\pi/2} \frac{\sin x \cos x}{1 + \sin x} dx$

$$= \left[ \sin x - \ln(\sin x + 1) \right]_0^{\pi/2}$$

$$= (1 - \ln 2) - (0)$$

$$= 1 - \ln 2$$

(c).

$$L(0) = 30$$

$$\therefore 30 = p + q$$

$$L'(0) = -14$$

Now,  $L'(x)$

$$= \frac{p}{3} e^{\frac{x}{3}} - \frac{2q}{3} e^{-\frac{2x}{3}}$$

$$\therefore -14 = \frac{p}{3} - \frac{2q}{3}$$

$$\therefore p - 2q = -42 \quad \text{--- (1)}$$

$$p + q = 30 \quad \text{--- (2)}$$

$$\Rightarrow p = 6, 6 + q = 30$$

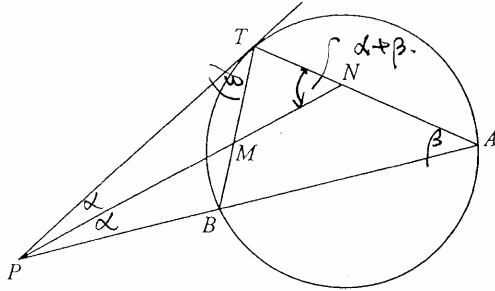
$$\therefore q = 24$$

$$\therefore L'(0) = -14 < 0$$

$$\text{and } L'(3) = 2e^{-16} e^{-2} > 0$$

$\therefore L(x_1)$  must be minimum for  $0 < x_1 < 3$ .

Question (7)



$$\angle PTB = \beta$$

$$\therefore \angle PTB = \beta$$

(alternate segment theorem.)

Now

$$\angle TNP = \alpha + \beta$$

(ext.  $\angle$  = sum of int. opp.  $\angle$ 's).

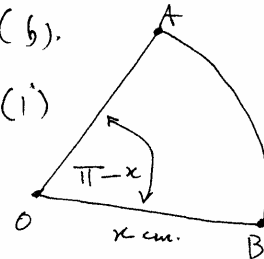
Similarly In  $\triangle TPM$ ,

$$\angle TMN = \alpha + \beta.$$

$\therefore \triangle TMN$  is isosceles.

(b).

(i)



$$P = 2x + x(\pi - x)$$

$$\therefore P = (\pi + 2)x - x^2$$

$$\frac{dP}{dx} = \pi + 2 - 2x$$

$$\frac{dP}{dx} = 0, \quad 2x = \pi + 2$$

$$\therefore x = \frac{\pi + 2}{2}$$

$$\frac{d^2P}{dx^2} = -2 < 0$$

$\therefore P$  is max when  $x = \frac{\pi + 2}{2}$

$$P_{\max} = \pi + 2 + \frac{\pi + 2}{2} \left( \frac{\pi}{2} - 1 \right)$$

$$= \pi + 2 + \left( \frac{\pi}{2} + 1 \right) \left( \frac{\pi}{2} - 1 \right)$$

$$= \pi + 2 + \frac{\pi^2}{4} - 1$$

$$= \frac{\pi^2}{4} + \pi + 1$$

$$= \frac{\pi^2 + 4\pi + 4}{4}$$

$$t(x) = \frac{x^2}{2} \sin(\pi - x)$$

$$\sin(\pi - x) = \sin x$$

$$\therefore t(x) = \frac{x^2 \sin x}{2}$$

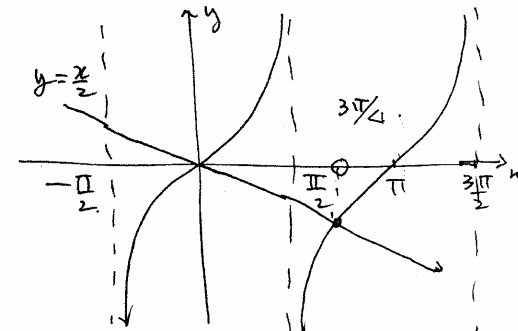
$$\frac{dt(x)}{dx} = x \sin x + \frac{x^2}{2} \cos x$$

$$\frac{dt(x)}{dx} = 0, \quad x \left( \sin x + \frac{x \cos x}{2} \right) = 0$$

$$\therefore \sin x = -\frac{x \cos x}{2}$$

$$\Rightarrow \tan x = -\frac{x}{2}$$

$$\therefore 2 \tan x = -x$$



$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= \frac{3\pi}{4} - \frac{-2 + \frac{3\pi}{4}}{1 + \frac{3\pi}{4}}$$

$$=$$